Week 1: Intro to demographic concepts

SOC6708 ADA

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Read in the data

All these data come from the UN World Population Prospects 2024.

Packages:

```
# in the root of the project, i.e., "/Hands_on/03_MonicaALabs/" folder where "03_MonicaALabs
library(tidyverse)
library(here)
library(readxl)
library(janitor)
```

Population data:

```
d_male <- read_xlsx("../data/WPP2024_POP_F01_2_POPULATION_SINGLE_AGE_MALE.xlsx", skip = 16, 
  mutate(Year = as.numeric(Year))
d_male$sex <- "Male"
d_male <- d_male |> drop_na(Year)
d_female <- read_xlsx("../data/WPP2024_POP_F01_3_POPULATION_SINGLE_AGE_FEMALE.xlsx", skip = 
  mutate(Year = as.numeric(Year))
d_female$sex <- "Female"
d_female <- d_female |> drop_na(Year)

d <- rbind(d_male, d_female)
rm(d_male, d_female)

d <- d  |>
  clean_names() |>
  select(region_subregion_country_or_area, iso3_alpha_code, year, x0:sex) |>
  rename(region = region_subregion_country_or_area) |>
```

```
  mutate(across(x0:x100, as.numeric))

head(d)
```

```
# A tibble: 6 x 105
  region iso3_alpha_code  year     x0     x1     x2     x3     x4     x5     x6
  <chr>  <chr>           <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
1 World  <NA>             1950 41603. 36997. 34245. 32359. 30007. 28065. 27335.
2 World  <NA>             1951 42615. 38588. 35577. 33456. 31876. 29664. 27829.
3 World  <NA>             1952 43984. 39615. 37182. 34798. 32980. 31545. 29427.
4 World  <NA>             1953 45183. 41001. 38243. 36421. 34326. 32646. 31320.
5 World  <NA>             1954 45934. 42199. 39641. 37489. 35955. 33999. 32422.
6 World  <NA>             1955 47059. 42972. 40848. 38889. 37023. 35636. 33784.
# i 95 more variables: x7 <dbl>, x8 <dbl>, x9 <dbl>, x10 <dbl>, x11 <dbl>,
#   x12 <dbl>, x13 <dbl>, x14 <dbl>, x15 <dbl>, x16 <dbl>, x17 <dbl>,
#   x18 <dbl>, x19 <dbl>, x20 <dbl>, x21 <dbl>, x22 <dbl>, x23 <dbl>,
#   x24 <dbl>, x25 <dbl>, x26 <dbl>, x27 <dbl>, x28 <dbl>, x29 <dbl>,
#   x30 <dbl>, x31 <dbl>, x32 <dbl>, x33 <dbl>, x34 <dbl>, x35 <dbl>,
#   x36 <dbl>, x37 <dbl>, x38 <dbl>, x39 <dbl>, x40 <dbl>, x41 <dbl>,
#   x42 <dbl>, x43 <dbl>, x44 <dbl>, x45 <dbl>, x46 <dbl>, x47 <dbl>, ...
```

Mortality data:

```
d_male <- read_xlsx(here("data/WPP2024_MORT_F01_2_DEATHS_SINGLE_AGE_MALE.xlsx"), skip = 16, c
  mutate(Year = as.numeric(Year))
d_male$sex <- "Male"
d_male <- d_male |> drop_na(Year)
d_female <- read_xlsx(here("data/WPP2024_MORT_F01_3_DEATHS_SINGLE_AGE_FEMALE.xlsx"), skip = 1
  mutate(Year = as.numeric(Year))
d_female$sex <- "Female"
d_female <- d_female |> drop_na(Year)

dm <- rbind(d_male, d_female)
rm(d_male, d_female)

dm <- dm  |>
  clean_names() |>
  select(region_subregion_country_or_area, iso3_alpha_code, year, x0:sex) |>
  rename(region = region_subregion_country_or_area) |>
  mutate(across(x0:x100, as.numeric))

head(dm)
```

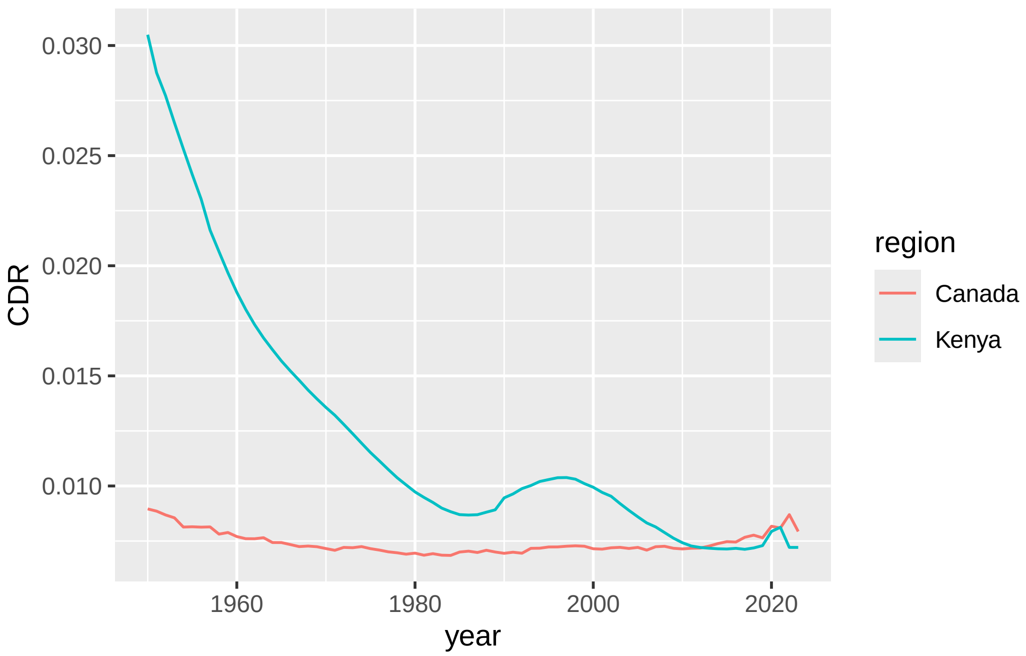
```
# A tibble: 6 x 105
  region iso3_alpha_code  year    x0    x1    x2    x3    x4    x5    x6    x7
  <chr>  <chr>           <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 World  <NA>             1950 6779. 1815.  967.  588.  380.  255.  183.  142.
2 World  <NA>             1951 6780. 1796.  968.  582.  382.  257.  181.  138.
3 World  <NA>             1952 6867. 1743.  943.  578.  375.  252.  176.  132.
4 World  <NA>             1953 6831. 1726.  933.  572.  379.  264.  191.  142.
5 World  <NA>             1954 6841. 1706.  928.  568.  375.  252.  180.  136.
6 World  <NA>             1955 6815. 1663.  917.  566.  373.  256.  181.  137.
# i 94 more variables: x8 <dbl>, x9 <dbl>, x10 <dbl>, x11 <dbl>, x12 <dbl>,
#   x13 <dbl>, x14 <dbl>, x15 <dbl>, x16 <dbl>, x17 <dbl>, x18 <dbl>,
#   x19 <dbl>, x20 <dbl>, x21 <dbl>, x22 <dbl>, x23 <dbl>, x24 <dbl>,
#   x25 <dbl>, x26 <dbl>, x27 <dbl>, x28 <dbl>, x29 <dbl>, x30 <dbl>,
#   x31 <dbl>, x32 <dbl>, x33 <dbl>, x34 <dbl>, x35 <dbl>, x36 <dbl>,
#   x37 <dbl>, x38 <dbl>, x39 <dbl>, x40 <dbl>, x41 <dbl>, x42 <dbl>,
#   x43 <dbl>, x44 <dbl>, x45 <dbl>, x46 <dbl>, x47 <dbl>, x48 <dbl>, ...
```

Crude Rates

Calculate the crude death rates for Kenya and Canada in 2023

```r
# get total populations across all age and sex
total_pops <- d  |>
  filter(region=="Kenya"|region=="Canada") |>
  pivot_longer(x0:x100, names_to = "age", values_to = "pop") |>
  mutate(age = as.numeric(str_remove(age, "x"))) |>
  group_by(region, year) |>
  summarize(total_pop = sum(pop))

# get total deaths

total_deaths <- dm  |>
  filter(region=="Kenya"|region=="Canada") |>
  pivot_longer(x0:x100, names_to = "age", values_to = "deaths") |>
  mutate(age = as.numeric(str_remove(age, "x"))) |>
  group_by(region, year) |>
  summarize(total_deaths = sum(deaths))

# join these

total_pops |>
```

```
  left_join(total_deaths) |>
  mutate(CDR = total_deaths/total_pop) |>
  ggplot(aes(year, CDR, color = region)) +
  geom_line()
```



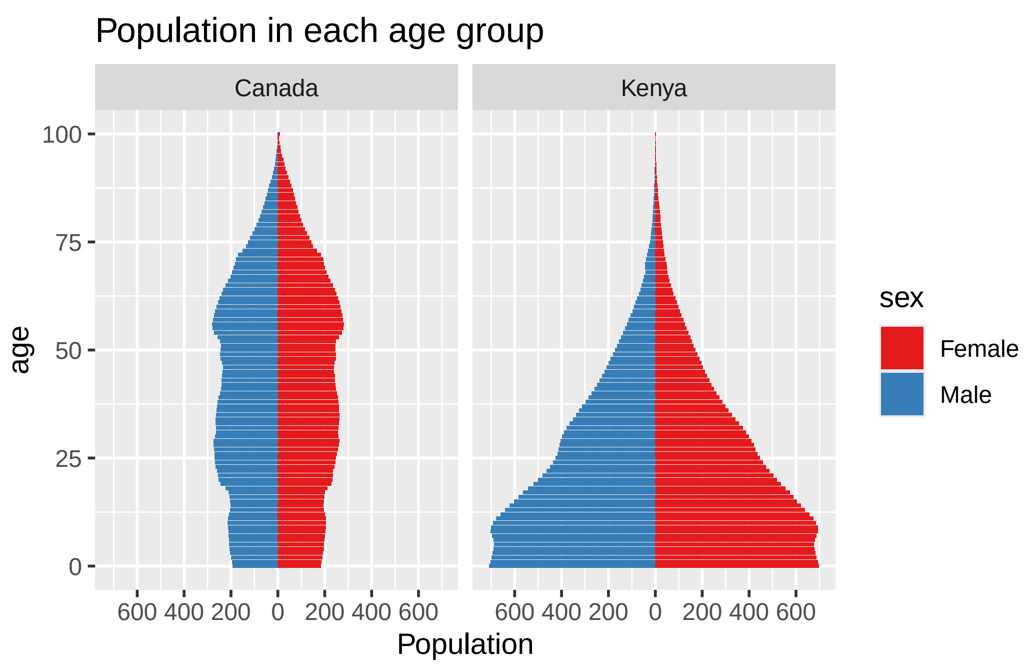

```
total_pops |>
  left_join(total_deaths) |>
  mutate(CDR = total_deaths/total_pop) |>
  filter(year==2023)
```

```
# A tibble: 2 x 5
# Groups:   region [2]
  region  year total_pop total_deaths     CDR
  <chr>  <dbl>     <dbl>        <dbl>   <dbl>
1 Canada  2023    39299.         312. 0.00793
2 Kenya   2023    55339.         399. 0.00721
```

Population Pyramids

Population age structures of Kenya and Canada in 2019:

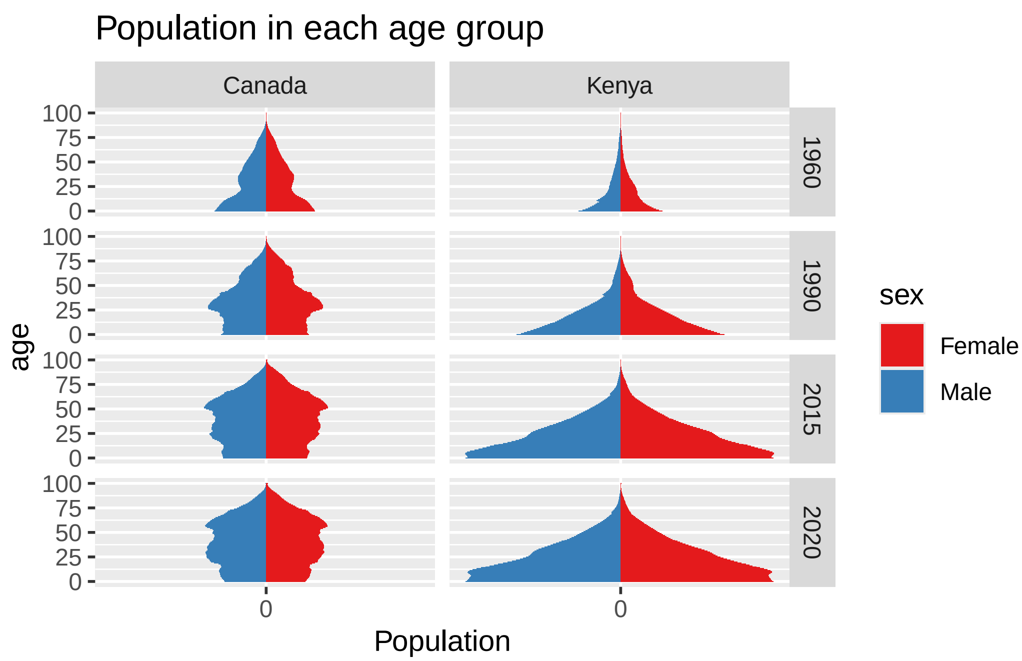
```r
d_long <- d  |>
  pivot_longer(x0:x100, names_to = "age", values_to = "pop") |>
  mutate(age = as.numeric(str_remove(age, "x")))

d_long|>
  filter(region=="Kenya"|region=="Canada", year == 2019)  |>
  mutate(population=ifelse(sex=="Male", -pop, pop)) |>
ggplot(aes(x = age, y = population, fill = sex)) +
  facet_wrap(~region)+
  geom_bar(stat="identity")+
  ggtitle("Population in each age group")+
  ylab("Population")+
  coord_flip() +
  scale_y_continuous(breaks = seq(-600, 600, 200),
                     labels = c(seq(600, 0, -200), seq(200, 600, 200))) +
  scale_fill_brewer(palette = "Set1")
```




```r
ggsave(here("plots", "CAN_KEN_pyramid.pdf"))
```

Change in age structures over time:

```
d_long |>
  filter(region=="Kenya"|region=="Canada", year%in% c(1960, 1990, 2015, 2020))  |>
  mutate(population=ifelse(sex=="Male", -pop, pop)) %>%
ggplot(aes(x = age, y = population, fill = sex)) +
  facet_grid(year~region)+
  geom_bar(stat="identity")+
  ggtitle("Population in each age group")+
  ylab("Population")+
  coord_flip() +
  scale_y_continuous(breaks = seq(-4000, 4000, 1000),
                     labels = c(seq(4000, 0, -1000), seq(1000, 4000, 1000))) +
  scale_fill_brewer(palette = "Set1")
```




```
ggsave(here("plots", "CAN_KEN_pyramid_time.pdf"))
```

Age-specific rates

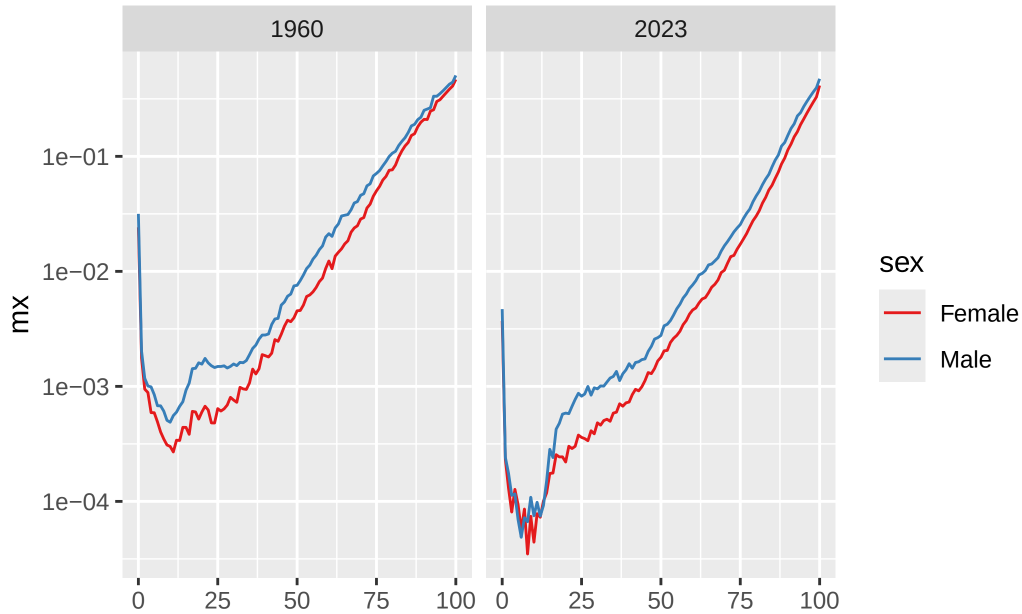
Create and plot age-specific mortality rates (across both sexes)

```r
pops <- d_long

dm_long <- dm  |>
  pivot_longer(x0:x100, names_to = "age", values_to = "deaths") |>
  mutate(age = as.numeric(str_remove(age, "x")))

# join these two tibbles and calculate rates

asmr <- d_long |>
  left_join(dm_long) |>
  mutate(mx = deaths/pop)

head(asmr)
```

```
# A tibble: 6 x 8
  region iso3_alpha_code  year sex     age    pop deaths      mx
  <chr>  <chr>           <dbl> <chr> <dbl>  <dbl>  <dbl>   <dbl>
1 World  <NA>             1950 Male      0 41603.  6779. 0.163
2 World  <NA>             1950 Male      1 36997.  1815. 0.0491
3 World  <NA>             1950 Male      2 34245.   967. 0.0282
4 World  <NA>             1950 Male      3 32359.   588. 0.0182
5 World  <NA>             1950 Male      4 30007.   380. 0.0127
6 World  <NA>             1950 Male      5 28065.   255. 0.00909
```

Mortality rates for Canada

```r
asmr |>
  filter(region == "Canada", year %in% c(1960, 2023)) |>
  ggplot(aes(age, mx, color = sex)) +
  geom_line()+
  scale_y_log10()+
  facet_wrap(~year)+
  labs(x ="Age- and sex-specific mortality rates for Canada")+
  scale_color_brewer(palette = "Set1")
```



Age− and sex−specific mortality rates for Canada

```
ggsave(here("plots", "CAN_mortality.pdf"), width = 6, height = 4)
```

Age-standardized rates

What would Kenya mortality look like in 2023 with Canada's age structure?

```
kenya_2023 <- asmr |>
  filter(year==2023, region=="Kenya")  |>
  rename(kpop = pop, kdeath = deaths, kmx = mx) |>
  select(-region)

canada_2023 <- asmr |>
  ungroup() |>
  filter(year==2023, region=="Canada")  |>
  rename(cpop = pop, cdeath = deaths, cmx = mx) |>
  select(-region, -age, -year, -sex, -iso3_alpha_code)

kc_2023 <- bind_cols(kenya_2023, canada_2023)

# now calculate age-standardized rates using canada's population
```

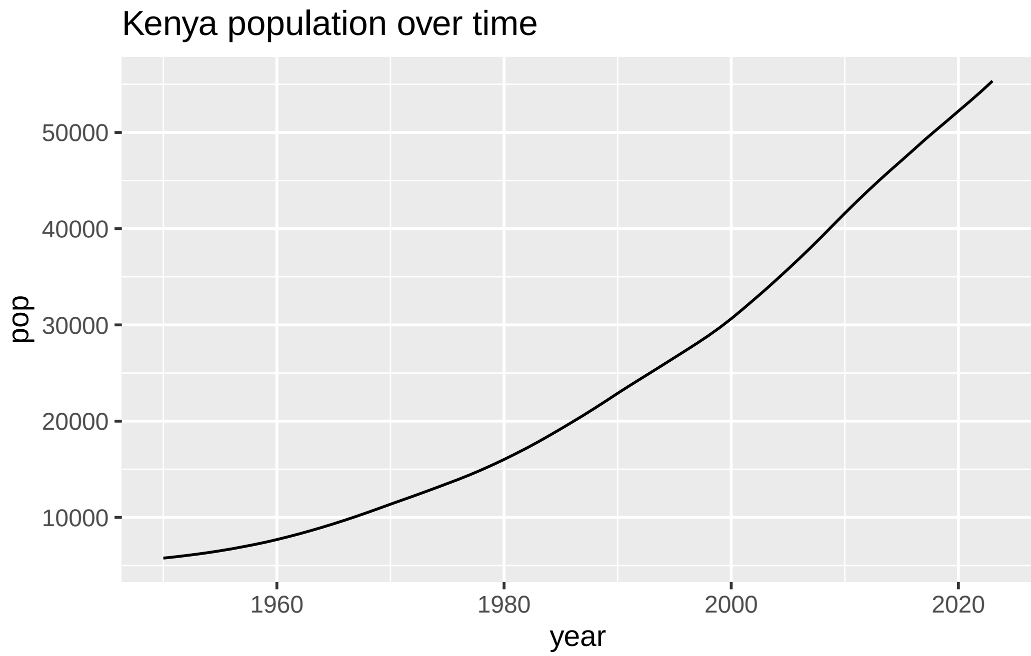
```r
kc_2023  |>
  mutate(std_deaths_kenya = cpop*kmx,
         std_deaths_canada = cpop*cmx) |>
  summarise(std_rate_kenya = sum(std_deaths_kenya)/sum(cpop),
            std_rate_canada = sum(std_deaths_canada)/sum(cpop))
```

```
# A tibble: 1 x 2
  std_rate_kenya std_rate_canada
           <dbl>           <dbl>
1         0.0223         0.00793
```

Population Growth

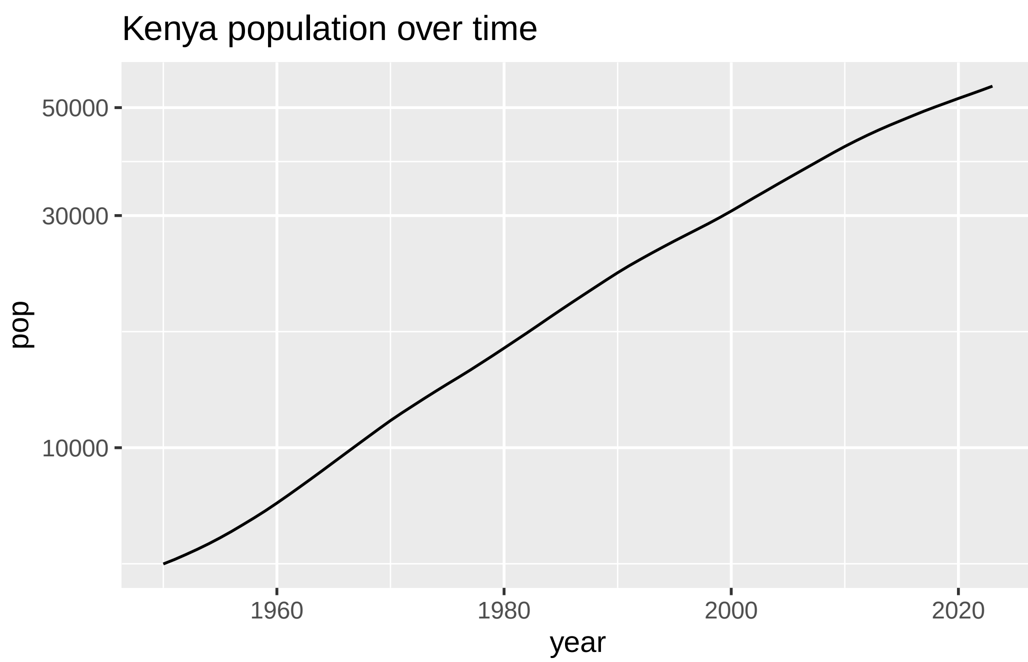
Plot Kenya total population over time

```r
d_long |>
  filter(region=="Kenya") |>
  group_by(year) |>
  summarize(pop = sum(pop)) |>
  ggplot(aes(year, pop)) +
  geom_line() +
  ggtitle("Kenya population over time")
```



What does the log look like?

```
d_long |>
  filter(region=="Kenya") |>
  group_by(year) |>
  summarize(pop = sum(pop)) |>
  ggplot(aes(year, pop)) +
  geom_line() +
  ggtitle("Kenya population over time")+
  scale_y_log10()
```




```
ggsave(here("plots", "KEN_pop.pdf"))
```

Pretty straight! Let's calculate the growth rate from 1950 to 2023 This is just the slope of the logged graph. It's about 3% per year.

```
d_long |>
  filter(region=="Kenya") |>
  group_by(year) |>
  summarize(pop = sum(pop)) |>
  mutate(log_pop = log(pop))|>
  summarise(growth_rate = (log_pop[year==2023] - log_pop[year==1950])/(2023-1950))
```

```
# A tibble: 1 x 1
  growth_rate
        <dbl>
1      0.0310
```

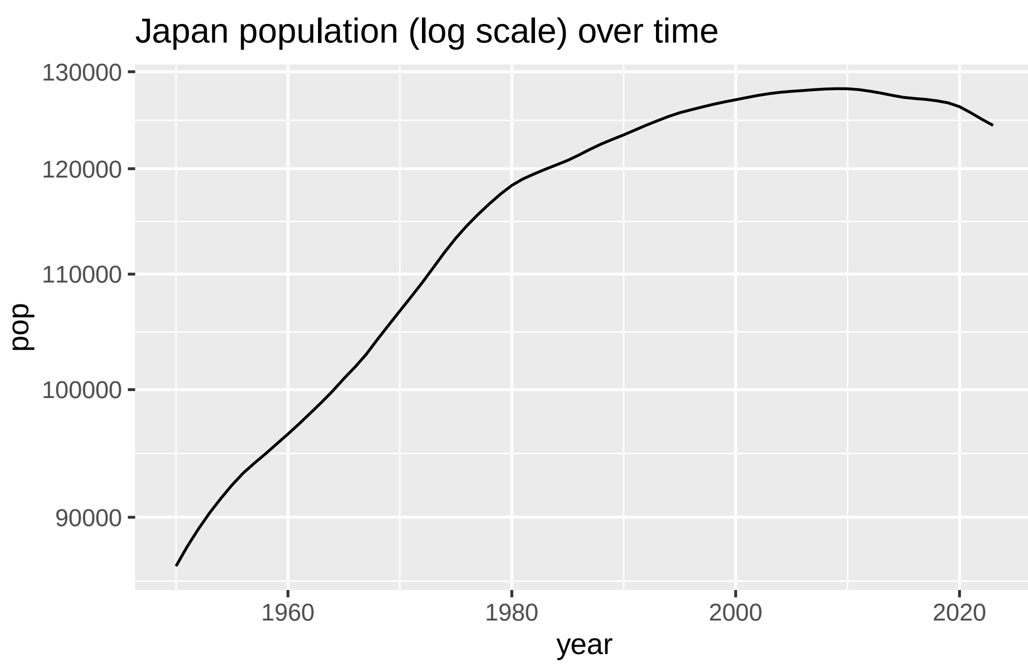
What about Canada? About half that.

```r
d_long |>
  filter(region=="Canada") |>
  group_by(year) |>
  summarize(pop = sum(pop)) |>
  mutate(log_pop = log(pop))|>
  summarise(growth_rate = (log_pop[year==2023] - log_pop[year==1950])/(2023-1950))
```

```
# A tibble: 1 x 1
  growth_rate
        <dbl>
1      0.0144
```

Some countries have stagnated:

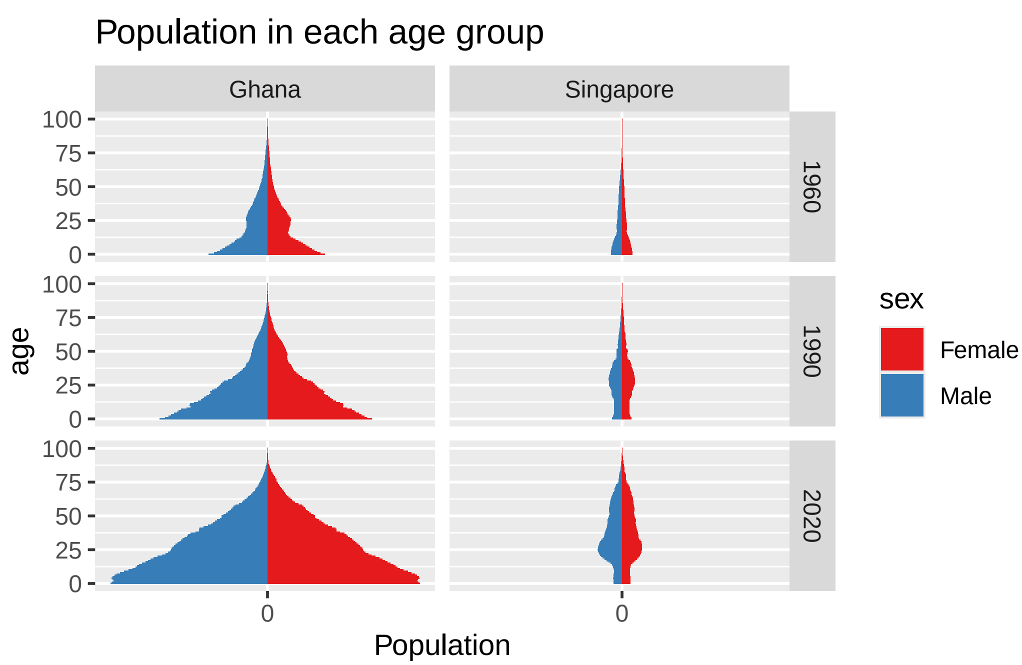
```r
d_long |>
  filter(region=="Japan") |>
  group_by(year) |>
  summarize(pop = sum(pop)) |>
  ggplot(aes(year, pop)) + geom_line()+
  scale_y_log10()+
  ggtitle("Japan population (log scale) over time")
```



Questions

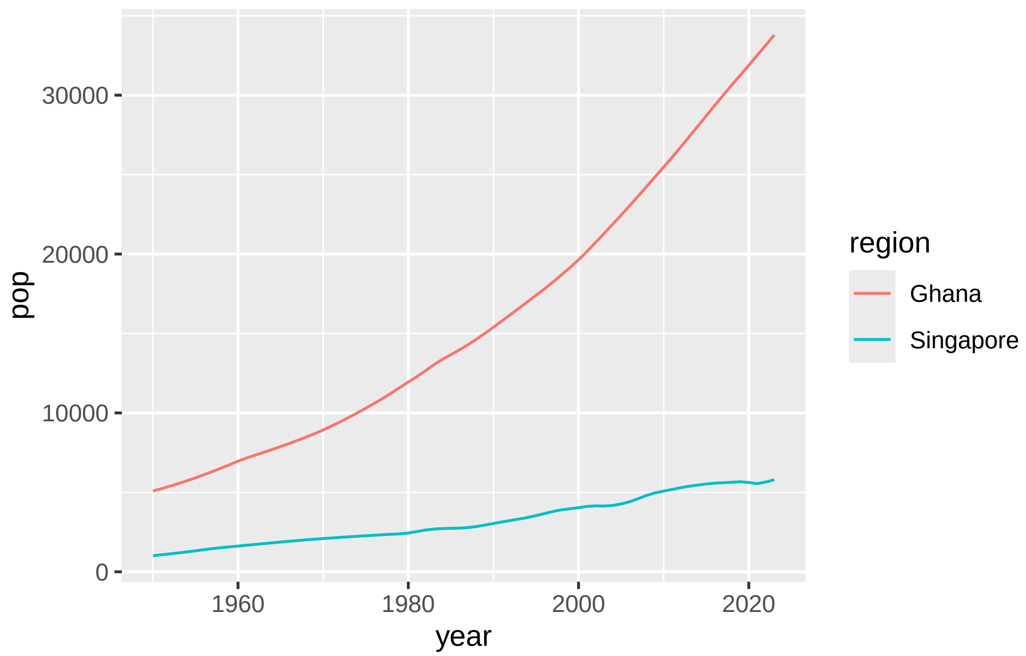
1. Pick two countries. Plot their population pyramids in 1960, 1990, and 2020.

```
d_long |>
  filter(region=="Singapore"|region=="Ghana", year%in% c(1960, 1990, 2020))  |>
  mutate(population=ifelse(sex=="Male", -pop, pop)) %>%
ggplot(aes(x = age, y = population, fill = sex)) +
  facet_grid(year~region)+
  geom_bar(stat="identity")+
  ggtitle("Population in each age group")+
  ylab("Population")+
  coord_flip() +
  scale_y_continuous(breaks = seq(-4000, 4000, 1000),
                     labels = c(seq(4000, 0, -1000), seq(1000, 4000, 1000))) +
  scale_fill_brewer(palette = "Set1")
```



2. Based on the shape and change in population pyramids in 1, do you think the population growth rate in each of your chosen countries is positive or negative in recent years?
3. Confirm your guesses in 2 by plotting population over time in both countries.

```
d_long |>
  filter(region=="Singapore"|region=="Ghana") |>
  group_by(region, year) |>
  summarize(pop = sum(pop)) |>
  ggplot(aes(year, pop, color = region))+
  geom_line()
```



4. Taking the US population in 2000 as the standard population, calculate the standardized mortality rates for US in 2023 and Australia in 2023. How much higher was the death rate in US compared to Australia?

```
std_pop <- d_long |>
  filter(region == "United States of America", year == 2000) |>
  select(sex, age, pop) |>
  rename(std_pop = pop)

asmr |>
  filter(region == "United States of America"| region == "Australia", year == 2023) |>
  left_join(std_pop) |>
  mutate(prod = mx*std_pop) |>
  group_by(region) |>
  summarize(asdr = sum(prod)/sum(std_pop))
```

```
# A tibble: 2 x 2
  region                     asdr
  <chr>                     <dbl>
1 Australia               0.00487
2 United States of America 0.00673
```

```r
asmr |>
  filter(region == "United States of America"| region == "Australia", year == 2023) |>
  left_join(std_pop) |>
  mutate(prod = mx*std_pop) |>
  group_by(region) |>
  summarize(asdr = sum(prod)/sum(std_pop)) |>
  summarize(ratio = asdr[region=="United States of America"]/asdr[region == "Australia"])
```

```
# A tibble: 1 x 1
  ratio
  <dbl>
1  1.38
```